Highlights

Modal Analysis of a Fluid-Filled Viscoelastic Oblate Spheroidal Shell: Implications for Airborne vs. Mechanical Low-Frequency Excitation

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- Fluid-filled shell model predicts abdominal resonance at 4–10 Hz
- Breathing mode at 2900 Hz; flexural modes match ISO 2631 data
- Airborne coupling penalised by $(ka)^n$; negligible at 120 dB SPL
- Mechanical vibration couples 1000–10000x more efficiently
- Coupling disparity explains WBV effects vs. absent airborne effects

Modal Analysis of a Fluid-Filled Viscoelastic Oblate Spheroidal Shell: Implications for Airborne vs. Mechanical Low-Frequency Excitation

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Abstract

The human abdomen is modelled as a fluid-filled viscoelastic oblate spheroidal shell to investigate its modal response under low-frequency excitation. Using Love–Kirchhoff thin-shell theory with Kelvin–Voigt viscoelasticity and proper fluid–structure coupling, we derive natural frequencies for both breathing ($n = 0$) and flexural ($n \geq 2$) mode families. The breathing mode, dominated by the fluid bulk modulus, occurs at approximately 2900 Hz and is irrelevant to infrasonic exposure. In contrast, flexural modes—in which the shell deforms without significant volume change and the enclosed fluid acts primarily as added mass—fall in the 4 Hz to 10 Hz range, consistent with abdominal resonance data reported under ISO 2631. A comprehensive parametric study over elastic modulus (0.05 MPa to 2.0 MPa), geometry, wall thickness, and intra-abdominal pressure demonstrates that the $n = 2$ flexural mode robustly occupies this frequency band for relaxed musculature, though the precise frequency is sensitive to tissue stiffness and boundary conditions.

We then compare two excitation pathways quantitatively. For airborne acoustic coupling, the long-wavelength condition ($ka \approx 0.017$ at 7 Hz) imposes a $(ka)^n$ penalty on flexural mode excitation, yielding displacements of order 0.1 µm at 120 dB SPL—below known mechanotransduction thresholds. For mechanical (whole-body) vibration, coupling is direct and produces displacements 10^3–10^4 times larger at occupational exposure levels. This disparity explains why whole-body vibration at 4 Hz to 8 Hz is associated with gastrointestinal

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effects in the epidemiological literature, whereas airborne infrasound at comparable frequencies is not. The analysis provides the first unified quantitative framework bridging these two excitation mechanisms through a common modal description.

Keywords: fluid-filled shell vibration, viscoelastic oblate spheroid, infrasound, whole-body vibration, acoustic–structural coupling, ISO 2631

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1. Introduction

Low-frequency acoustic and vibrational exposures are ubiquitous in occupational and environmental settings—from wind turbines and heavy machinery to transport vehicles and industrial plant. The human body is known to exhibit resonance phenomena in the 4 Hz to 8 Hz range under whole-body vibration, with the abdominal viscera identified as the primary resonant subsystem [1, 2].

A persistent claim in popular culture holds that a specific infrasonic frequency in the 5 Hz to 9 Hz range—the so-called "brown note"—can cause involuntary loss of bowel control through resonance of the abdominal cavity. Despite widespread repetition, no rigorous scientific study has confirmed this claim, and it has been explicitly debunked in controlled experiments [3].

However, the complete absence of a *quantitative mechanistic analysis* leaves the question incompletely resolved. No study has:

1. derived abdominal cavity eigenfrequencies from first-principles shell theory with proper fluid–structure coupling;
2. quantified the airborne acoustic coupling coefficient for flexural modes in the long-wavelength limit; or
3. compared airborne and mechanical excitation pathways using a common modal framework.

The present work addresses all three gaps. We model the abdomen as a fluid-filled viscoelastic oblate spheroidal shell and derive natural frequencies for both breathing ($n = 0$) and flexural ($n \geq 2$) mode families. We then

compare the efficiency of airborne acoustic coupling (subject to a $(ka)^n$ penalty) against mechanical whole-body vibration coupling (which bypasses this penalty entirely).

The central finding is that the coupling disparity between these two pathways—a factor of 10^3–10^4—quantitatively explains why whole-body vibration at 4 Hz to 8 Hz produces gastrointestinal effects while airborne infrasound at comparable frequencies does not.

The remainder of this paper is organised as follows. Section 2 develops the mathematical formulation for the fluid-filled viscoelastic shell, including both breathing and flexural mode families and the two coupling pathways. Section 3 presents a comprehensive parametric study and validates the model against ISO 2631 data. Section 4 quantifies the airborne–mechanical coupling disparity. Section 5 interprets the results and discusses limitations, and Section 6 summarises the conclusions.

2. Mathematical Formulation

2.1. Model Geometry and Assumptions

The human abdominal cavity is idealised as an oblate spheroidal shell of semi-major axes a (equatorial, in the lateral and cranio-caudal directions) and semi-minor axis c (antero-posterior), with $c < a$. The shell has uniform wall thickness h and is completely filled with an inviscid, irrotational fluid of density ρ_f and bulk modulus K_f. Following standard practice in fluid-filled shell analysis [4, 5], an equivalent-sphere radius is defined via

$$R_{\mathrm{eq}} = (a^2 c)^{1/3}, \tag{1}$$

which preserves the cavity volume, $V = \frac{4}{3}\pi a^2 c$. This approximation is exact for a sphere ($c = a$) and introduces errors of order $(1-c/a)^2$ for the eigenfrequencies at moderate aspect ratios ($c/a \gtrsim 0.5$). Table 1 lists the baseline geometric and material parameters with physiological justification.

The thin-shell assumption ($h/R_{\mathrm{eq}} \ll 1$) is satisfied: for $h = 10\,\mathrm{mm}$ and $R_{\mathrm{eq}} \approx 160\,\mathrm{mm}$, $h/R \approx 0.06$. The Love–Kirchhoff hypothesis (normals remain straight and normal to the deformed mid-surface) is therefore applicable.

Table 1: Baseline model parameters with physiological justification.

Parameter	Symbol	Value	Unit	Source
Semi-major axis	a	0.18	m	Anthropometric mean
Semi-minor axis	c	0.12	m	$c/a \approx 0.67$
Wall thickness	h	0.010	m	Composite wall
Elastic modulus	E	0.1	MPa	Relaxed passive tissue
Poisson's ratio	ν	0.45	—	Nearly incompressible
Wall density	ρ_w	1050	$\mathrm{kg/m^3}$	Soft tissue
Fluid density	ρ_f	1040	$\mathrm{kg/m^3}$	Water-like
Fluid bulk modulus	K_f	2.2	GPa	Water at 37°C
Intra-abdominal pressure	P_{iap}	1000	Pa	Resting supine
Loss tangent	η	0.30	—	Viscoelastic tissue

2.2. Shell Constitutive Model

The abdominal wall is modelled as a Kelvin–Voigt viscoelastic material with complex modulus

$$E^* = E(1 + i\eta), \tag{2}$$

where η is the loss tangent. For small damping, the quality factor is $Q = 1/\eta$ and the viscous damping ratio is $\zeta = \eta/2 = 1/(2Q)$. The flexural rigidity of the shell is

$$D = \frac{Eh^3}{12(1-\nu^2)}. \tag{3}$$

Published values of η for abdominal soft tissue range from 0.15 to 0.40 at low frequencies [6], corresponding to $Q = 2.5$–6.7. The baseline value $\eta = 0.30$ ($Q = 3.3$) represents a central estimate.

2.3. Breathing Mode ($n = 0$)

The breathing mode is a spatially uniform radial oscillation in which the shell displaces by ξ at every point. The associated volumetric strain is $\Delta V/V = 3\xi/R$, which compresses the contained fluid. Two restoring forces act on the shell:

Membrane stiffness.. For the uniform expansion of a thin spherical shell,

$$k_{\text{shell}} = \frac{2Eh}{R^2(1-\nu)}. \tag{4}$$

Fluid volumetric stiffness.. The pressure increment due to fluid compression is $\Delta p = K_f \Delta V/V = 3K_f\xi/R$. The restoring force per unit displacement per unit area is

$$k_{\text{fluid}} = \frac{3K_f}{R}. \tag{5}$$

With $K_f = 2.2\,\text{GPa}$ and $R = 0.157\,\text{m}$, $k_{\text{fluid}} = 4.2 \times 10^{10}\,\text{Pa/m}$, which exceeds the membrane stiffness ($k_{\text{shell}} \approx 2.4 \times 10^5\,\text{Pa/m}$) by five orders of magnitude. *The breathing mode is entirely dominated by the fluid bulk modulus.*

The effective mass per unit area includes the shell wall and the fluid added mass for the $n = 0$ mode:

$$m_{\text{eff}}^{(0)} = \rho_w h + \rho_f R. \tag{6}$$

The breathing mode frequency is therefore

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{k_{\text{shell}} + k_{\text{fluid}}}{\rho_w h + \rho_f R}} \approx 2900\ \text{Hz}, \tag{7}$$

which lies in the kilohertz range and is irrelevant to infrasonic exposure. This result is robust to material property variations because $k_{\text{fluid}} \gg k_{\text{shell}}$ for any realistic shell stiffness.

2.4. Flexural Modes ($n \geq 2$)

Flexural modes involve shape changes (oblate–prolate oscillation for $n = 2$, trilobed for $n = 3$, etc.) that preserve the enclosed volume to leading order. The fluid therefore does *not* act as a volumetric spring; it contributes only added mass.

Following Lamb [5] and Junger & Feit [4], the flexural mode of order n has three contributions to the restoring force per unit area:

Bending stiffness..

$$K_{\text{bend}} = \frac{n(n-1)(n+2)^2 D}{R^4}. \tag{8}$$

Membrane stiffness..

$$K_{\text{memb}} = \frac{Eh}{R^2}\,\lambda_n, \qquad \lambda_n = \frac{n^2+n-2+2\nu}{n^2+n+1-\nu}. \tag{9}$$

Pre-stress stiffness.. The static intra-abdominal pressure P produces a pre-stress in the shell membrane that stiffens flexural modes:

$$K_P = \frac{P}{R}(n-1)(n+2). \tag{10}$$

The fluid added mass per unit area for mode n is

$$m_{\text{add}}^{(n)} = \frac{\rho_f R}{n}, \tag{11}$$

giving a total effective mass of

$$m_{\text{eff}}^{(n)} = \rho_w h + \frac{\rho_f R}{n}. \tag{12}$$

The natural frequency of flexural mode n is

$$\omega_n^2 = \frac{K_{\text{bend}} + K_{\text{memb}} + K_P}{m_{\text{eff}}^{(n)}}, \qquad f_n = \frac{\omega_n}{2\pi}. \tag{13}$$

For the baseline parameters (Table 1), $f_2 = 4.4\,\text{Hz}$, $f_3 = 6.7\,\text{Hz}$, and $f_4 = 9.4\,\text{Hz}$. These lie squarely in the 4–10 Hz range reported for abdominal resonance in ISO 2631 [1] and in the experimental studies of Kitazaki & Griffin [7] and Mansfield [8].

2.5. Airborne Acoustic Coupling

We now quantify the excitation of flexural modes by an incident airborne acoustic field. At 7 Hz, the acoustic wavelength in air is $\lambda \approx 49\,\text{m}$, so the dimensionless wavenumber is

$$ka = \frac{2\pi f R}{c_{\text{air}}} \approx 0.017 \ll 1. \tag{14}$$

The body is deep in the sub-wavelength (Rayleigh scattering) regime.

In a multipole expansion of the incident plane wave scattered by a sphere, the n-th order component of the pressure field scales as [4]:

$$p_{\text{eff}}^{(n)} = p_{\text{inc}} \times (ka)^n. \tag{15}$$

The breathing mode ($n = 0$) sees the full incident pressure, but flexural modes ($n \geq 2$) see only the higher-order spatial gradients, which are vanishingly small when $ka \ll 1$. For $n = 2$:

$$p_{\text{eff}}^{(2)} = p_{\text{inc}} \times (ka)^2 \approx p_{\text{inc}} \times 2.8 \times 10^{-4}. \tag{16}$$

The displacement at resonance ($\omega = \omega_n$) is

$$\xi_{\text{air}} = \frac{p_{\text{eff}}^{(n)}}{K_{\text{total}}^{(n)}} Q, \tag{17}$$

where $K_{\text{total}}^{(n)} = K_{\text{bend}} + K_{\text{memb}} + K_P$. At 120 dB SPL ($p_{\text{inc}} = 20\,\text{Pa}$), equation (17) gives $\xi_{\text{air}} \approx 0.14\,\mu\text{m}$ for $n = 2$ — well below the 0.5–2.0 μm activation threshold of mechanosensitive ion channels [9].

2.6. Mechanical (Whole-Body Vibration) Coupling

Whole-body vibration (WBV) transmits through the skeleton and acts as *base excitation* of the abdominal shell. Unlike airborne acoustic coupling, there is no impedance mismatch and no $(ka)^n$ penalty. The vibration is transmitted at full amplitude through the rigid skeletal frame.

For harmonic base acceleration $a_{\text{base}}(t) = a_0 \cos(\omega t)$, the base displacement amplitude is

$$x_{\text{base}} = \frac{a_{\text{rms}}\sqrt{2}}{\omega^2}. \tag{18}$$

The relative displacement between the shell and the skeleton is given by the base-excitation frequency response function:

$$H_{\text{rel}}(\omega) = \frac{r^2}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}, \qquad r = \frac{\omega}{\omega_n}, \tag{19}$$

so that

$$\xi_{\text{mech}} = x_{\text{base}} \times H_{\text{rel}}. \tag{20}$$

At the EU Whole-Body Vibration Directive exposure limit of $a_{\mathrm{rms}} = 1.15\,\mathrm{m/s^2}$ and at the $n = 2$ resonance frequency:

$$\xi_{\mathrm{mech}} \approx 7{,}000\ \mu\mathrm{m} = 7\ \mathrm{mm}, \tag{21}$$

which exceeds the mechanotransduction threshold by more than three orders of magnitude.

2.7. Coupling Ratio

The ratio of mechanical to airborne displacement at the same excitation conditions provides a dimensionless measure of the coupling disparity:

$$\mathcal{R} = \frac{\xi_{\mathrm{mech}}}{\xi_{\mathrm{air}}} \sim 10^3\text{–}10^4. \tag{22}$$

This ratio depends primarily on ka and the excitation level, and is *robust* to uncertainties in material properties and boundary conditions because both pathways excite the same modal structure. The coupling ratio is the central result of this work: it provides a quantitative explanation for why whole-body vibration at 4–8 Hz is associated with gastrointestinal effects in the epidemiological literature [2], whereas airborne infrasound at comparable frequencies is not.

3. Parametric Study

3.1. Elastic Modulus Sensitivity

The $n = 2$ flexural mode frequency depends strongly on the wall elastic modulus E. Table 2 shows the predicted frequency for a free shell across the physiologically relevant range.

The 4 Hz to 8 Hz ISO 2631 abdominal resonance range requires $E < 0.5\,\mathrm{MPa}$, corresponding to relaxed, passive musculature. Active muscle bracing increases E by a factor of 10–100, moving the resonance well above the infrasound range.

Table 2: Flexural mode $n = 2$ frequency vs. elastic modulus ($a = 0.18\,\mathrm{m}$, $c = 0.12\,\mathrm{m}$, $h = 10\,\mathrm{mm}$).

E (MPa)	f_2 (Hz)	In ISO 2631 range?
0.05	3.6	Marginal
0.10	4.4	Yes
0.20	5.6	Yes
0.50	8.3	Yes
1.00	11.5	No
2.00	16.1	No

3.2. Multi-Parameter Sensitivity

A full factorial analysis over 486 parameter combinations (6 values of E, 3 semi-major axes, 3 aspect ratios, 3 wall thicknesses, 3 loss tangents) reveals that 37% of configurations place the $n = 2$ mode in the 5 Hz to 10 Hz range and 41% in the broader 4 Hz to 8 Hz ISO 2631 range. The dominant parameter is E, followed by geometry and intra-abdominal pressure.

3.3. Uncertainty Quantification

A Monte Carlo simulation with $N = 10{,}000$ samples propagates parameter uncertainties through the model. Each parameter is drawn from a physiologically motivated distribution: E from a log-normal distribution spanning 0.05 MPa to 2.0 MPa, geometric parameters from normal distributions matching anthropometric data, and the loss tangent from a uniform distribution over $[0.10,\ 0.40]$.

The resulting distribution of f_2 has a median of 7.5 Hz with a 90% credible interval of 3.3 Hz to 15.5 Hz (Table 3). The distribution is positively skewed, reflecting the log-normal distribution of E.

Sobol sensitivity analysis reveals that the elastic modulus accounts for 86% of the total output variance (total-order index $S_T = 0.86$). All other parameters contribute less than 10% individually. This result has important experimental

Table 3: Monte Carlo uncertainty quantification results ($N = 10{,}000$).

Quantity	Median	90% CI	Units
f_2	7.5	[3.3, 15.5]	Hz
ξ_{air} (120 dB)	0.07	[0.01, 0.4]	μm
ξ_{mech} (0.5 m/s^2)	600	[100, 3000]	μm
$\mathcal{R}$	10^4	[10^3, 10^5]	—
SPL for PIEZO	135	[128, 139]	dB

implications: any future validation effort should prioritise *in vivo* measurement of abdominal wall stiffness, as this single parameter dominates the uncertainty budget.

3.4. Multi-Layer Wall Model

The abdominal wall comprises approximately five distinct tissue layers: peritoneum, transversalis fascia, internal oblique muscle, external oblique muscle, subcutaneous fat, and skin. Using classical laminate theory, the effective composite modulus for a relaxed wall is $E_{\text{eff}} \approx 0.075\,\text{MPa}$ ($h_{\text{total}} = 25.1\,\text{mm}$), giving $f_2 = 4.6\,\text{Hz}$—within 4% of the homogeneous model prediction. Muscle tension is confirmed as the dominant factor: tensed muscle ($E_{\text{eff}} = 2.7\,\text{MPa}$) shifts f_2 to 23.1 Hz.

3.5. Boundary Condition Effects

The free-shell model provides a lower bound on frequency. Partial constraints (spine, pelvis, ribs) raise the $n = 2$ frequency by an estimated factor of 1.3–3.0×, depending on the angular extent of the constrained boundary. Even with a 2× constraint multiplier, the mode remains within the ISO 2631 range for soft tissue.

3.6. ISO 2631 Validation

Table 4 compares model predictions against published abdominal resonance data.

Table 4: Model predictions vs. published abdominal resonance frequencies.

Body type	f_2 (Hz)	ISO 2631 range?
Lean ($E = 0.2$, $a = 15$ cm)	6.3	Yes
Average ($E = 0.1$, $a = 18$ cm)	4.0	Yes
Large ($E = 0.08$, $a = 20$ cm)	3.1	Marginal

4. Airborne vs. Mechanical Coupling

This section presents the central novel result of this work: a quantitative comparison of airborne acoustic and mechanical (whole-body vibration) coupling to the abdominal flexural modes derived in §2.

4.1. Airborne Acoustic Pathway

Table 5 summarises the predicted airborne displacement at resonance for the $n = 2$ flexural mode across a range of sound pressure levels.

Table 5: Airborne displacement at $n = 2$ resonance ($f_2 = 4.4\,\mathrm{Hz}$, $Q = 3.3$, $E = 0.1\,\mathrm{MPa}$).

SPL (dB)	p_{inc} (Pa)	p_{eff} (Pa)	ξ (μm)	PIEZO?
100	2.0	3.2×10^{-4}	0.014	No
110	6.3	1.0×10^{-3}	0.045	No
120	20.0	3.2×10^{-3}	0.14	No
130	63.2	1.0×10^{-2}	0.45	No
140	200.0	3.2×10^{-2}	1.44	Marginal

At 120 dB SPL (the threshold of pain), the predicted wall displacement is 0.14 µm—well below the 0.5–2.0 µm activation range of mechanosensitive PIEZO channels. Reaching the 1 µm threshold requires approximately 137 dB, a level associated with close proximity to jet engines and acoustic weapons.

4.2. Mechanical (Whole-Body Vibration) Pathway

In contrast, mechanical vibration transmitted through the skeleton couples directly to the shell without the $(ka)^n$ penalty. Table 6 shows the predicted

relative displacement for the $n = 2$ mode under whole-body vibration.

Table 6: Mechanical coupling at $n = 2$ resonance ($f_2 = 4.4\,\mathrm{Hz}$, $Q = 3.3$).

a_{rms} (m/s^2)	x_{base} (μm)	ξ_{rel} (μm)	PIEZO?
0.1	187	623	Yes
0.5 (EU action)	935	3115	Yes
1.0	1869	6230	Yes
1.15 (EU limit)	2150	7165	Yes

At the EU Whole-Body Vibration Directive action value of $0.5\,\mathrm{m/s^2}$, the predicted displacement is 3115 µm—exceeding the PIEZO threshold by more than three orders of magnitude.

4.3. Coupling Ratio

The ratio of mechanical to airborne displacement defines the coupling disparity:

$$\mathcal{R} = \frac{\xi_{\mathrm{mech}}}{\xi_{\mathrm{air}}} \sim 10^3\text{–}10^4. \tag{23}$$

This ratio depends primarily on $(ka)^{-n}$ and the base-excitation amplification H_{rel}, and is robust to uncertainties in material properties because both pathways excite the same modes.

4.4. Energy Budget

To verify self-consistency, we compute the absorption cross-section using reciprocity:

$$\sigma_{\mathrm{abs}} = \frac{(2n+1)\lambda^2}{4\pi}\,\frac{\Gamma_{\mathrm{rad}}}{\Gamma_{\mathrm{total}}}, \tag{24}$$

where Γ_{rad} is the radiation damping width and $\Gamma_{\mathrm{total}} = \Gamma_{\mathrm{rad}} + \Gamma_{\mathrm{struct}}$. For the $n = 2$ mode in air, $\zeta_{\mathrm{rad}} \approx 10^{-15}$ whereas $\zeta_{\mathrm{struct}} = 0.15$, giving an absorption efficiency of order 10^{-14}. The shell is effectively transparent to airborne infrasound.

4.5. Comparison with Empirical Evidence

The model predictions are consistent with two independent bodies of evidence:

- **WBV studies**: ISO 2631 documents gastrointestinal symptoms at 4–8 Hz under mechanical vibration [1].
- **Airborne infrasound studies**: No gastrointestinal effects have been observed at SPL $\leq 150\,\mathrm{dB}$ [3].

The coupling disparity provides a quantitative explanation for this apparent contradiction: the two bodies of literature address different excitation mechanisms with fundamentally different coupling efficiencies.

5. Discussion

5.1. The Coupling Disparity

The central result of this study is the 10^3–10^4 factor separating mechanical from airborne coupling efficiency (Tables 5–6). The physical origin of this disparity lies in two compounding effects: an impedance mismatch and a geometric filtering penalty.

First, the air–tissue impedance mismatch is severe. The specific acoustic impedance of air is $Z_{\mathrm{air}} = \rho_{\mathrm{air}} c_{\mathrm{air}} \approx 420\,\mathrm{Pa\,s/m}$, while that of soft tissue is $Z_{\mathrm{tissue}} \approx 1.6 \times 10^6\,\mathrm{Pa\,s/m}$, giving a pressure transmission coefficient $T = 2Z_{\mathrm{air}}/(Z_{\mathrm{air}} + Z_{\mathrm{tissue}}) \approx 5 \times 10^{-4}$. At the body surface, more than 99.9 % of the incident acoustic energy is reflected.

Second, even the transmitted pressure must couple to the spatial pattern of a flexural mode. In the long-wavelength (Rayleigh) limit, $ka \ll 1$, the n-th order component of the incident plane wave scales as $(ka)^n$ [4]. At 7 Hz with $R_{\mathrm{eq}} = 0.133\,\mathrm{m}$, we obtain $ka = 2\pi f R/c_{\mathrm{air}} \approx 0.017$ and therefore $(ka)^2 \approx 2.8 \times 10^{-4}$ for the $n = 2$ flexural mode (equation 16). Physically, the incident wavelength ($\lambda \approx 49\,\mathrm{m}$) is nearly 370 times the body diameter; the pressure field is almost

perfectly uniform over the body surface, and the quadrupolar ($n = 2$) pressure gradient that could drive an oblate–prolate deformation is vanishingly small.

The energy budget analysis (§4) confirms this picture from an independent perspective. The radiation damping ratio in air is $\zeta_{\mathrm{rad}} \approx 10^{-15}$, whereas the structural damping ratio is $\zeta_{\mathrm{struct}} = 0.15$, so the absorption efficiency $\zeta_{\mathrm{rad}}/\zeta_{\mathrm{total}} \sim 10^{-14}$. The shell is, for practical purposes, acoustically transparent at infrasonic frequencies. By contrast, mechanical vibration transmitted through the skeleton bypasses both barriers entirely: there is no impedance mismatch (bone couples directly to the peritoneal wall) and no geometric filtering (the base-excitation drives the $n = 2$ mode at full amplitude via equation 19).

This disparity resolves an apparent paradox in the literature. Whole-body vibration at 4 Hz to 8 Hz is well documented to cause gastrointestinal distress—including nausea, abdominal pain, and in extreme cases, incontinence [2, 8]—whereas airborne infrasound studies at levels up to 150 dB report no comparable effects [10, 3]. The present modal analysis shows that both observations are consistent with the *same* underlying resonance. The two excitation pathways differ not in the resonance they address, but in the efficiency with which they address it.

5.2. Implications for the Brown Note Hypothesis

The results permit a nuanced evaluation of the so-called "brown note" hypothesis. The hypothesis is *half-correct*: abdominal resonance in the 4 Hz to 10 Hz band is physically real, well-supported by the parametric study (§3), and consistent with ISO 2631 transmissibility data [1, 7]. The flexural $n = 2$ mode at 4.4 Hz (for $E = 0.1$ MPa, relaxed musculature) falls squarely within the frequency band where occupational exposure studies report gastrointestinal symptoms.

Where the hypothesis fails is in the *excitation pathway*. At 120 dB SPL—the threshold of pain and well above any environmental or entertainment exposure—the predicted airborne wall displacement is 0.14 µm (Table 5), a factor of 4–14 below the 0.5 µm to 2.0 µm activation threshold of mechanosensitive PIEZO1/2

ion channels [9]. Reaching the 1 µm threshold requires approximately 137 dB, a level associated only with close proximity to jet engines and directed acoustic devices. Even at this extreme, the displacement remains three orders of magnitude below what routine whole-body vibration ($a_{\mathrm{rms}} = 0.5\,\mathrm{m/s^2}$) produces (Table 6).

This framing is, however, *constructive* rather than merely debunking. The model provides the first quantitative explanation for *why* whole-body vibration at 4 Hz to 8 Hz is so effective at producing gastrointestinal effects: it excites a bona fide flexural resonance of the fluid-filled abdominal cavity, producing millimetre-scale tissue displacements that far exceed mechanotransduction thresholds. The epidemiological pattern documented by ISO 2631—peak symptom incidence at 4 Hz to 8 Hz, with a strong dependence on acceleration amplitude—is precisely what a resonance-driven mechanism predicts.

5.3. Alternative Coupling Pathways

The whole-cavity resonance model of §2 treats the abdominal wall as a homogeneous fluid-filled shell. In reality, two additional mechanisms provide alternative pathways for airborne sound to produce tissue displacement: (i) compression of intestinal gas pockets, and (ii) orifice coupling through the upper gastrointestinal tract. We consider each in turn.

5.3.1. Intestinal gas pocket response

The human intestine normally contains 100 mL to 200 mL of gas distributed in pockets with effective radii ranging from 1 cm to 10 cm. Unlike the surrounding tissue, these gas volumes are compressible, and they respond to incident pressure by expanding and contracting their bounding walls.

The classical Minnaert frequency of a spherical gas bubble of radius a in a fluid of density ρ_f is [5]

$$f_M = \frac{1}{2\pi a}\sqrt{\frac{3\gamma P_0}{\rho_f}}, \tag{25}$$

where $\gamma = 1.4$ is the ratio of specific heats and P_0 is the ambient pressure. For physiological pocket radii of 1 cm to 10 cm, this yields Minnaert frequencies

in the range 33 Hz to 330 Hz—well above the infrasonic band of interest. At 7 Hz, the frequency ratio $f/f_M \approx 0.02$–0.2, so gas pockets are driven far below resonance and experience no resonant amplification ($H \approx 1.0$).

Nevertheless, in the sub-resonant regime the gas pocket wall displacement is governed by the quasi-static compliance of the trapped gas:

$$\xi_{\text{gas}} \approx \frac{p_{\text{inc}}\, a}{3\gamma P_0}, \tag{26}$$

which scales linearly with pocket radius and is independent of frequency. At 120 dB SPL ($p_{\text{inc}} = 20\,\text{Pa}$) and $a = 2\,\text{cm}$, this yields $\xi_{\text{gas}} \approx 0.94\,\mu\text{m}$—within the PIEZO activation range and roughly seven times the whole-cavity airborne displacement of 0.14 µm at the same SPL. For larger pockets ($a = 5\,\text{cm}$), the displacement rises to approximately 2.4 µm, exceeding the threshold with a modest margin.

Crucially, at 7 Hz the acoustic wavelength (49 m) vastly exceeds the body diameter, so the pressure field inside the body is spatially uniform to within $(ka)^2$ corrections. The gas pockets therefore experience the full incident pressure, without the geometric filtering penalty that suppresses whole-cavity flexural excitation. The mechanism operates as a local pressure-to-displacement transducer embedded within the tissue, rather than as a global shell resonance.

This gas pocket pathway has two important implications. First, it may explain the well-documented inter-individual variability in infrasound sensitivity: bowel gas content varies enormously (50 mL to 500 mL) with diet, gut motility, and microbiome composition, and individuals with greater gas volume would present more sites for pressure–displacement conversion. Second, the mechanism does not require resonance and operates at any frequency in the sub-wavelength regime, broadening the range of potentially relevant exposures.

5.3.2. Orifice coupling through the upper GI tract

The mouth and nose provide a low-impedance air pathway to the pharynx and—when the esophagus is patent—to the stomach. At 7 Hz, the total path length from mouth to stomach ($L \approx 40\,\text{cm}$) satisfies $kL \approx 0.05 \ll 1$, so pressure

is transmitted through the air column with negligible viscothermal attenuation.

With the mouth open and esophagus patent, the pressure reaching the gastric air bubble is approximately equal to the incident pressure, representing a transmission ratio of order unity compared with the $(ka)^2 \approx 2.8 \times 10^{-4}$ penalty for body-surface coupling—an enhancement factor of roughly 2,400×. Compression of the gastric gas bubble then produces wall displacements by the same mechanism described above for intestinal pockets.

However, this pathway is severely limited by anatomy. The esophagus is a collapsible muscular tube that is closed at rest; it opens only during swallowing (approximately 0.5 s to 1.0 s per event), belching, or pathological gastroesophageal reflux. At rest with the mouth open, the air column terminates at the upper esophageal sphincter, and pressure reaches only the supraglottic space. The orifice pathway is therefore *intermittent*, dependent on the state of the upper GI sphincters, and cannot be relied upon as a continuous coupling mechanism.

5.3.3. Assessment of alternative pathways

Table 7 summarises the predicted displacement for each coupling mechanism at 120 dB SPL and 7 Hz.

Table 7: Comparison of coupling mechanisms at 120 dB SPL, 7 Hz.

Mechanism	ξ (μm)	PIEZO?
Cavity resonance (n=2, airborne)	0.14	No
Gas pocket $a = 2$ cm (airborne)	0.94	Marginal
Gas pocket $a = 5$ cm (airborne)	2.4	Yes
Orifice (open mouth → stomach)	∼0.5	Marginal
WBV at 0.5 m/s^2 (mech.)	3,115	Yes
WBV at 1.15 m/s^2 (mech.)	7,165	Yes

Even with the most favourable alternative pathway (a large, 5 cm-radius gas pocket at 120 dB), the airborne displacement remains two to three orders

of magnitude below what mechanical coupling achieves at the EU Whole-Body Vibration Directive action value of $0.5\,\mathrm{m/s^2}$. The alternative pathways therefore do not alter the principal conclusion of this work—that airborne infrasound is an ineffective excitation route compared with mechanical vibration—but they provide a physically grounded explanation for why rare, sporadic gastrointestinal responses to extreme infrasound (>120 dB) are not entirely implausible, and why such responses exhibit large inter-individual variability.

5.4. Clinical and Occupational Relevance

The coupling disparity has direct implications for occupational exposure standards. ISO 2631-1 specifies frequency-weighted acceleration limits for whole-body vibration, with the highest sensitivity weighting in the 4 Hz to 8 Hz band—precisely the range where our model predicts the $n = 2$ flexural resonance [1]. The EU Physical Agents (Vibration) Directive 2002/44/EC codifies two thresholds: a daily exposure action value of $0.5\,\mathrm{m/s^2}$ $A(8)$ and an exposure limit value of $1.15\,\mathrm{m/s^2}$ $A(8)$. At the action value, our model predicts a relative abdominal wall displacement of 3.1 mm at resonance (Table 6)—more than 1,500× the PIEZO activation threshold. The present analysis therefore provides a *mechanistic rationale* for why these particular frequency bands and amplitude thresholds were empirically selected: they correspond to conditions under which flexural resonance of the abdominal cavity produces tissue strains large enough to activate mechanotransduction pathways.

For airborne infrasound, ISO 7196 defines G-weighting but prescribes no dose–response limits. Our analysis supports this regulatory asymmetry: at environmental and occupational SPL (≤ 120 dB), airborne coupling to abdominal flexural modes is negligible, and exposure limits based on abdominal resonance effects are unnecessary at these levels.

A potentially valuable extension of this framework is toward personalised vibration exposure assessment. The parametric study (§3) shows that the $n = 2$ eigenfrequency depends sensitively on body habitus: larger abdominal radius (higher BMI) reduces the resonance frequency, while greater muscular tone

(higher effective E) increases it. In principle, imaging-derived geometric parameters [6] combined with the analytical expressions of §2 could yield subject-specific resonance frequency estimates, enabling targeted vibration exposure limits for workers whose body habitus places their resonance frequency in a particularly hazardous band.

5.5. Limitations

We identify the following limitations, grouped by their likely effect on the conclusions.

1. **Linear theory.** The Love–Kirchhoff shell formulation assumes small displacements. Under mechanical vibration at occupational amplitudes, the predicted displacements (several millimetres) approach the wall thickness ($h \approx 10\,\mathrm{mm}$), where geometric nonlinearity and strain-stiffening of biological tissue become relevant [11]. Nonlinear effects would *limit* the actual displacement below the linear prediction, thereby reducing the mechanical coupling estimate somewhat but not the coupling *ratio*, because the airborne displacement is already deep in the linear regime. The principal conclusion is therefore robust to this limitation.
2. **Homogeneous, isotropic wall.** The abdominal wall is a layered anisotropic composite comprising four muscle layers, fascia, subcutaneous fat, and skin. Our laminate analysis (not reproduced here) shows that homogenisation introduces less than 5 % error in the effective bending stiffness. The isotropic assumption is a first-order approximation; anisotropy would split the mode degeneracies of the equivalent sphere but would not change the order of magnitude of the eigenfrequencies or the coupling ratio.
3. **Free-shell boundary conditions.** The abdomen is partially constrained by the spine posteriorly, the pelvis inferiorly, and the rib cage superiorly. The free-shell model provides a *lower bound* on flexural mode frequencies. Partial clamping is estimated to raise the $n = 2$ frequency by a factor of 2–3×, which could shift the predicted resonance above the 4 Hz to 8 Hz

ISO 2631 band for stiffer tissue configurations. This limitation *weakens* the agreement with ISO 2631 data for stiff-tissue subjects but does not affect the coupling ratio, because both excitation pathways address the same modal structure. Finite-element validation with realistic anatomical constraints is a priority for future work.

4. **Thin-shell approximation.** The wall-thickness-to-radius ratio ($h/R \approx 0.06$) lies near the conventional upper bound of thin-shell theory ($h/R < 0.05$). First-order shear-deformation (Mindlin–Reissner) corrections would reduce the $n = 2$ frequency by approximately 4 % [12], a negligible effect compared with the material-property uncertainty. This limitation is unlikely to affect any conclusions.
5. **Equivalent-sphere geometry.** The oblate spheroidal abdomen is mapped to an equivalent sphere via $R_{\mathrm{eq}} = (a^2c)^{1/3}$. This preserves volume and leading-order curvature but suppresses the equatorial–polar mode splitting that occurs in an oblate spheroid [13]. For a typical aspect ratio $c/a = 0.67$, the splitting is estimated at $\sim$33 %, which could shift individual mode frequencies but would not change the coupling ratio or the qualitative separation between breathing and flexural mode families.
6. **No internal organ structure.** Visceral organs are homogenised into a single inviscid fluid. Solid inclusions (liver, spleen) would locally stiffen the enclosed medium and introduce additional scattering, potentially shifting eigenfrequencies. This limitation is difficult to assess without full three-dimensional modelling, but it is expected to redistribute modal energy rather than fundamentally alter the coupling disparity.
7. **Gas pocket model simplifications.** The gas pocket analysis (§5.3) treats intestinal gas as spherical bubbles in an infinite homogeneous fluid, neglecting the cylindrical geometry of the gut lumen, viscoelastic wall loading, and the surrounding organ structure. These simplifications likely *overestimate* gas pocket displacement, because intestinal wall mass loading and stiffness would reduce the effective compliance. A coupled gas–tissue finite-element model is needed to refine these estimates.

8. **Mechanotransduction thresholds.** The 0.5 µm to 2.0 µm PIEZO activation range [9] derives from cell-culture experiments and may not directly translate to tissue-scale effects, where strain concentrations around cells are modulated by the extracellular matrix. This limitation introduces uncertainty in the absolute threshold but does not affect the *relative* comparison between airborne and mechanical coupling.

In summary, the identified limitations affect the absolute values of predicted frequencies and displacements—in several cases substantially—but none alter the central finding that airborne acoustic coupling to abdominal flexural modes is 10^3–10^4 times weaker than mechanical coupling under occupationally relevant conditions.

6. Conclusions

A fluid-filled viscoelastic oblate spheroidal shell model of the human abdomen reveals two distinct mode families with fundamentally different physics. The breathing mode ($n = 0$), governed by the fluid bulk modulus, occurs at approximately 2900 Hz and is irrelevant to infrasonic exposure. Flexural modes ($n \geq 2$), in which the fluid acts only as added mass, fall in the 4 Hz to 10 Hz range for relaxed musculature, consistent with ISO 2631 whole-body vibration data.

The central finding is a 10^3–10^4 coupling disparity between mechanical and airborne excitation of these flexural modes. This disparity, arising from the $(ka)^n$ long-wavelength penalty on acoustic coupling, provides a quantitative mechanistic explanation for an otherwise puzzling asymmetry in the empirical literature: mechanical vibration at 4 Hz to 8 Hz produces gastrointestinal effects, while airborne infrasound at comparable frequencies does not. The "brown note" hypothesis is thus physically grounded—the abdominal resonance is real—but the postulated *airborne* excitation pathway is energetically negligible.

Future work should validate the boundary condition assumptions through finite element analysis, extend the model to include gas pocket resonance in

the bowel, and investigate the non-linear regime relevant to high-amplitude mechanical vibration.

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Data availability

All analytical models and computational scripts used in this study are available in the accompanying repository.

CRediT authorship contribution statement

Jon: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – Original Draft, Writing – Review & Editing, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.